Nifty Options, India VIX and a VIX-Timed Hedge

Volatility dynamics, the 4 June 2024 election, and a backtest of protection bought in calm markets and sold into fear

FDRM course project · Group 8, IIT Jodhpur

Four numbers that sum up the study

2.2 pts

Average amount by which India VIX exceeded the volatility that followed (VIX higher on 78% of days)

9% → 30%

Nifty's annualised volatility when VIX is below 13 vs above 20

4.2×

Jump in a June 22,500 Nifty put on 4 June 2024, election result day

−22%

Worst fall with the VIX-timed hedge, vs −38% for unhedged Nifty (2020–2026)

Fear is overpriced most of the time and most overpriced just after a shock. So buy protection when VIX is calm and sell it into the spike.

Real prices, modelled option premiums

Real market data

- Nifty 50 daily closes, Sep 2007 to Apr 2026
- India VIX daily closes, Jan 2020 to Aug 2026
- Common sample: 1,536 trading days, Jan 2020 to Apr 2026
- Checked against NSE: on 4 Jun 2024 Nifty closed at 21,884.50 and VIX at 26.75

Modelled option premiums

- Black-Scholes-Merton, risk-free 6.5%, dividend yield 1.3%
- Implied vol = India VIX + 0.6 vol pt per 1% out of the money
- Bought at +0.5 and sold at −0.5 vol pt to cover bid/ask
- Historical NSE option chains were not available

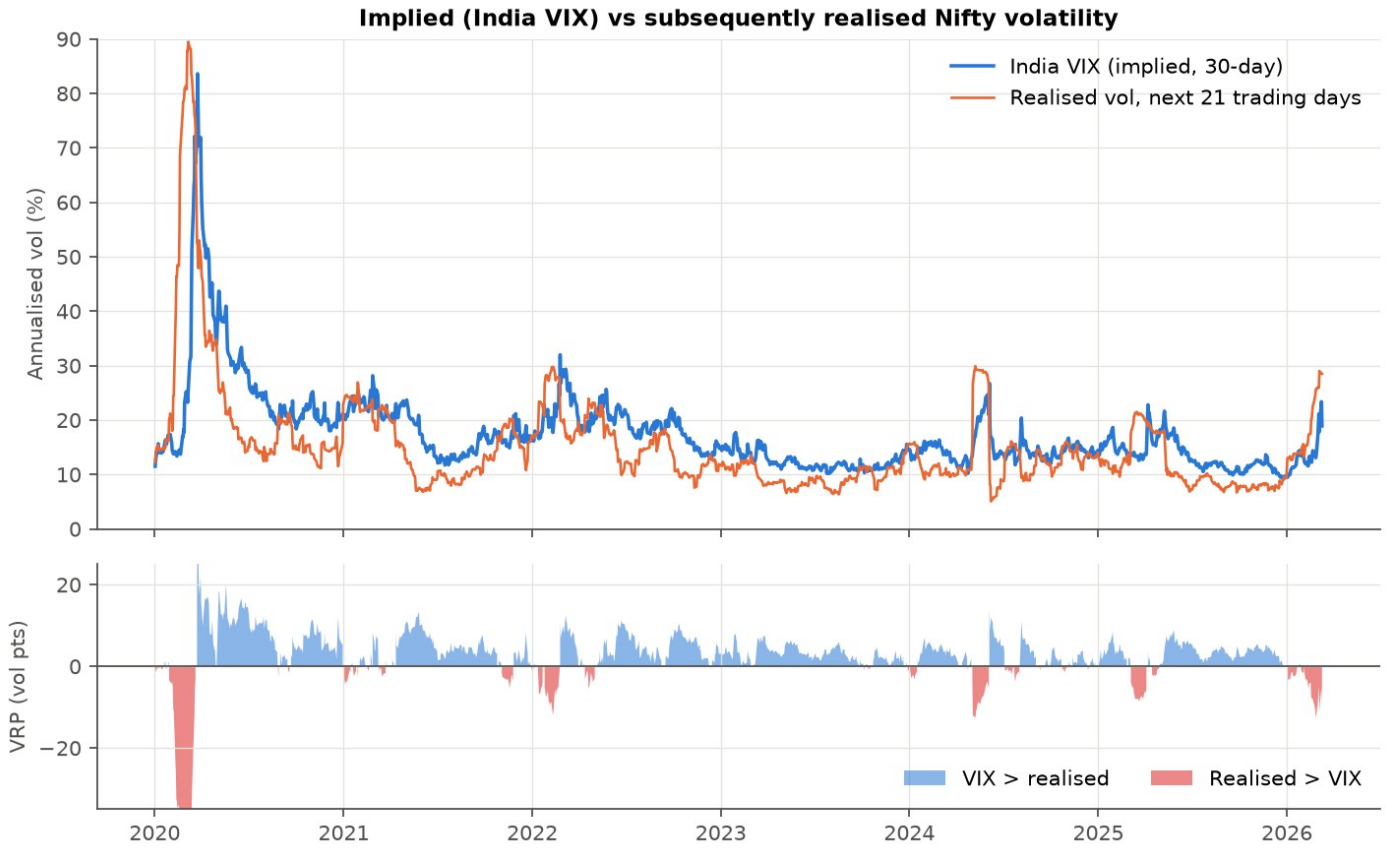
VIX regimes used throughout

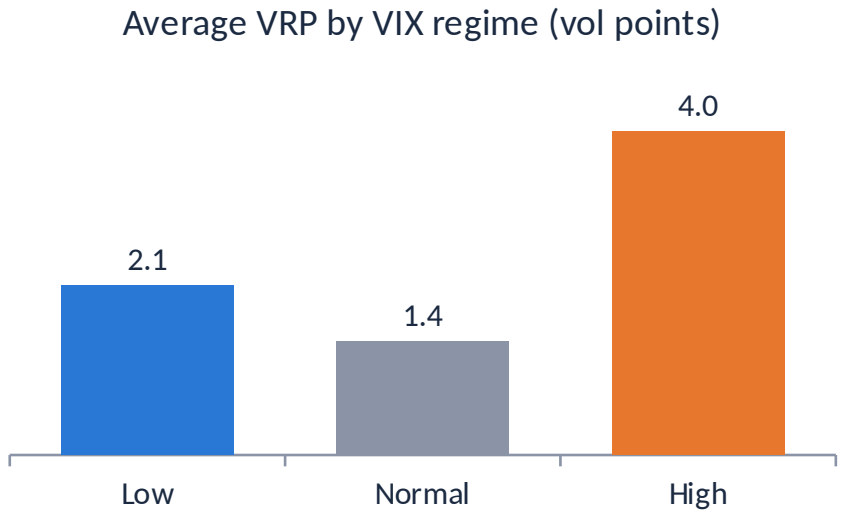
Low: VIX below 13
27% of days

Normal: 13 to 20
48% of days

High: above 20
25% of days

VIX usually overstates future volatility





- VRP = VIX minus realised vol over the next 21 days: +2.2 pts on average (t = 2.8)
- Largest after shocks (4.0 pts): sell protection then, don't buy it
- VIX tracks the past 21 days (corr 0.86) more than the next 21 (0.61)

How India VIX is built

1

Near and next-month options

OTM puts below the forward, OTM calls above it, at mid-quotes

2

Variance per expiry

Weights each option by $\Delta K/K^2$, so low strikes count most

3

Interpolate to 30 days

Blend the two expiries to a constant 30-calendar-day horizon

4

Scale to an index

India VIX = 100 × √(30-day variance), in annualised percent

$$\sigma^2 = (2/T) \cdot \Sigma\, (\Delta K_i / K_i^2) \cdot e^{(RT)} \cdot Q(K_i) \; - \; (1/T) \cdot (F / K_0 - 1)^2$$

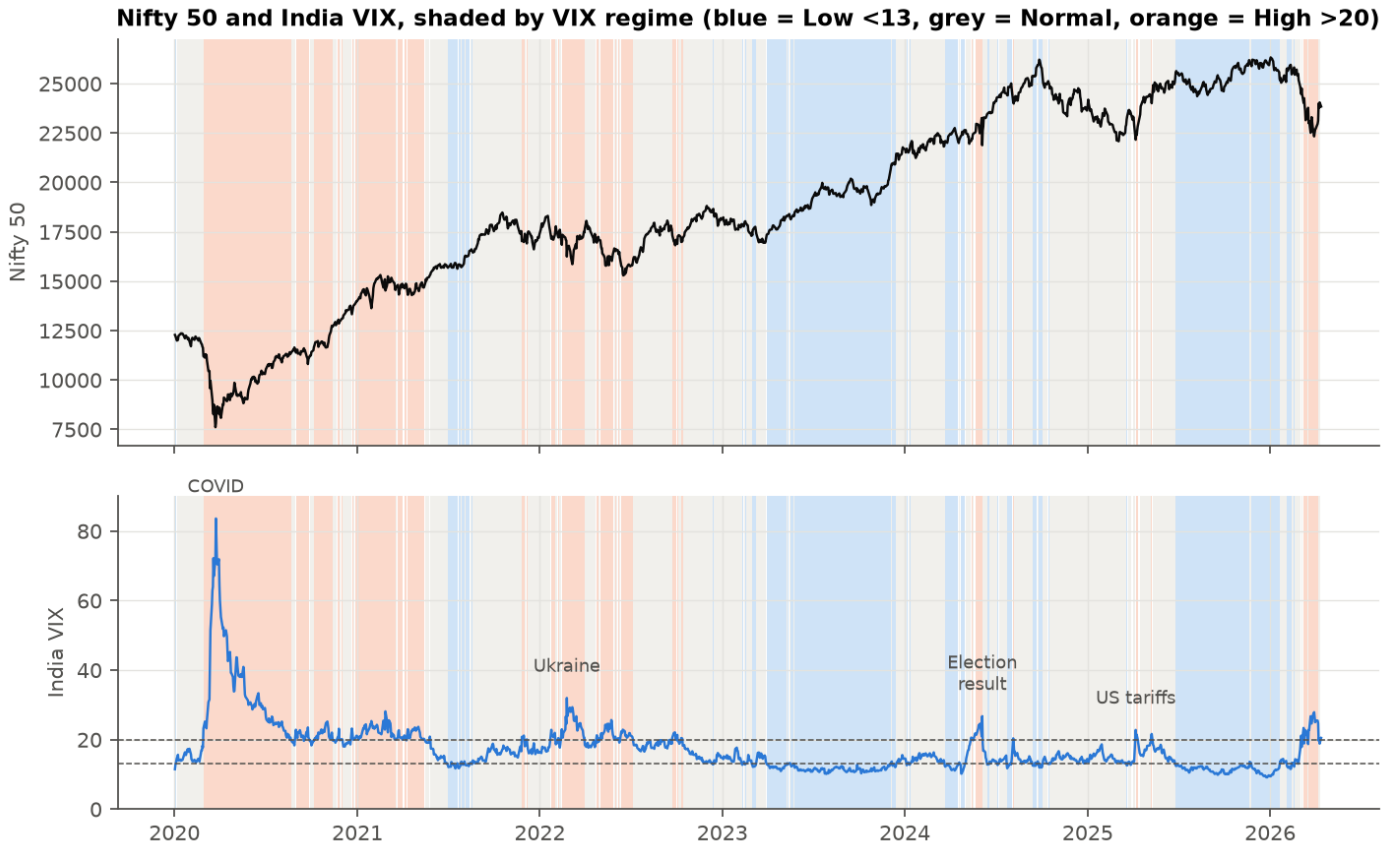
15.0

Formula on a synthetic chain with a flat 15% smile: it recovers the input exactly

16.1

Same 15% at-the-money vol with put skew: VIX also prices crash fear

India VIX as a fear gauge: three traits



1

Rises when Nifty falls

+2.3% on an average down day, −1.9% on an up day (correlation −0.56)

2

Persists, then mean-reverts

Regimes last weeks to months, but every spike returned to 12–16

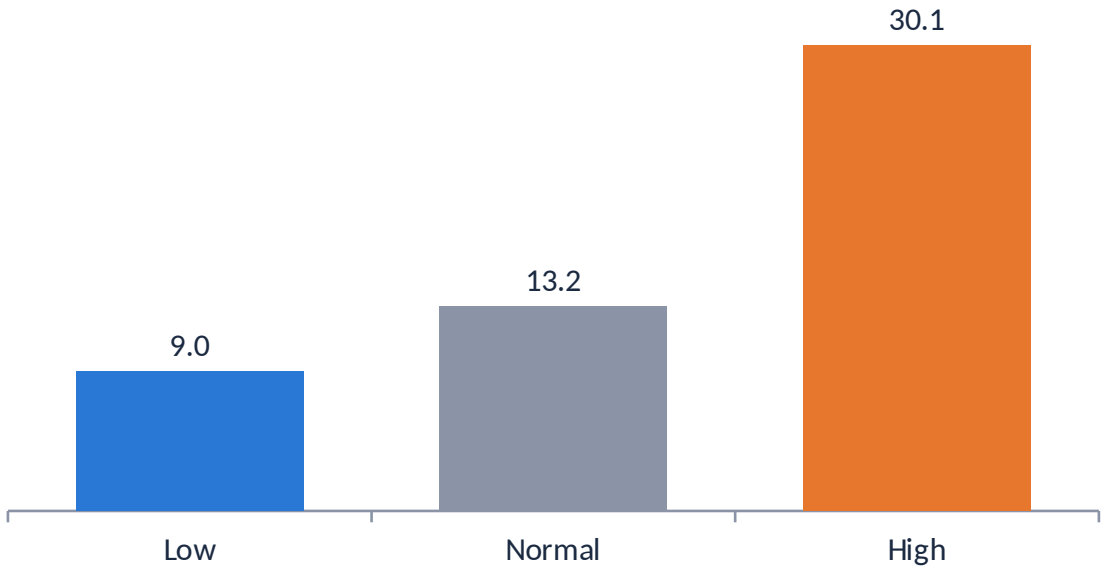
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Prices known events early

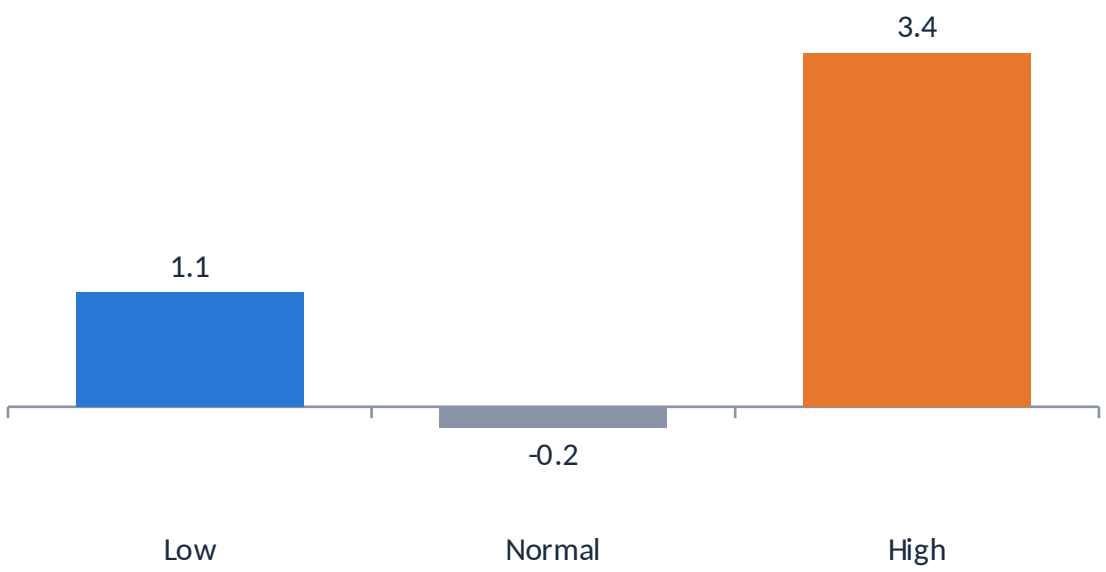
VIX climbed for six weeks before the 2024 result while Nifty was flat

VIX regimes separate risk, not return

Annualised Nifty volatility by VIX regime (%)

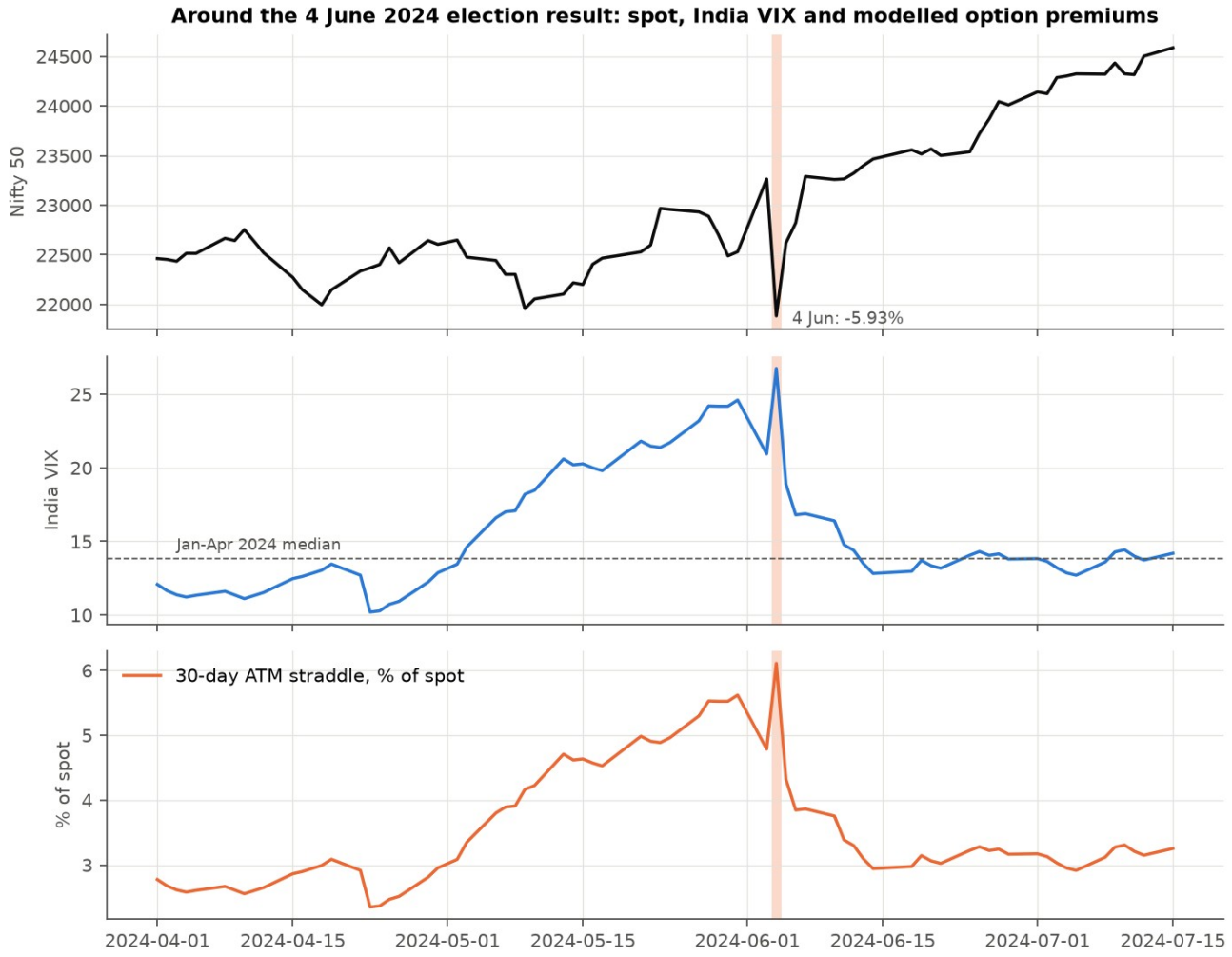


Average Nifty return over the next 21 days (%)



- Volatility differs by regime (Levene $p < 0.001$), but same-day returns do not (Kruskal-Wallis $p = 0.66$)
- High VIX was followed by rebounds: +0.18% next-21-day return per VIX point ($t = 4.3$), mostly the 2020 recovery
- Worst 63-day outcomes began in the Normal regime (5th percentile −14.6%), so protection must be in place before VIX rises

4 June 2024: build-up, spike, collapse



1

Late Apr to 31 May: run-up

VIX 10.2 → 24.6 with Nifty flat; 30-day straddle 2.4% → 5.6% of spot

2

3 June: exit polls

Nifty +3.25%, VIX −15%: protection was cheapest just before the shock

3

4 June: result

Nifty −5.93% (−8.5% intraday), VIX 26.75; June 22,500 put up 4.2×

4

5 to 7 June: vol crush

VIX −29% next day as Nifty rebounded 3.4%; straddle back to 3.9%

Priced the size, not the direction

Day	Nifty move	VIX	Straddle, % of spot	Move vs implied
Election result (Jun 2024)	−5.93%	20.9 → 26.8	4.8 → 6.1	4.6×
Exit polls (Jun 2024)	+3.25%	24.6 → 20.9	5.6 → 4.8	2.1×
US tariffs (Apr 2025)	−3.24%	13.8 → 22.8	3.2 → 5.2	3.8×
Yen carry (Aug 2024)	−2.68%	14.3 → 20.4	3.3 → 4.7	3.0×
Ukraine (Feb 2022)	−4.78%	24.5 → 32.0	5.6 → 7.3	3.2×
COVID (Mar 2020)	−12.98%	67.1 → 72.0	15.3 → 16.4	3.3×
Normal day (median)	±0.47%	14.7	3.4	0.5×

Straddle = 30-day at-the-money, modelled from VIX. Move vs implied = absolute move ÷ (previous day's VIX ÷ √252).

5.8%

Move implied by the 31 May VIX (normal VIX 14.4)

5.9%

Actual fall on 4 June

4.6×

Largest move vs implied in the sample

Proposed hedge: buy in calm, sell into fear

1

Buy

VIX below 13 at month start: buy a 3-month put 5% below spot on the whole portfolio

2

Roll

While VIX stays below 13, swap it each month for a fresh 3-month put

3

Wait

VIX 13 or above: buy nothing new and hold the existing put to expiry

4

Sell half

First close with VIX at 22 or more: sell 50% of the puts

5

Sell the rest

VIX 20% below its post-spike peak: sell the rest and reinvest in Nifty

Options are cheapest in calm markets

Low regime: VIX averaged 11.7 and realised vol 9.5%

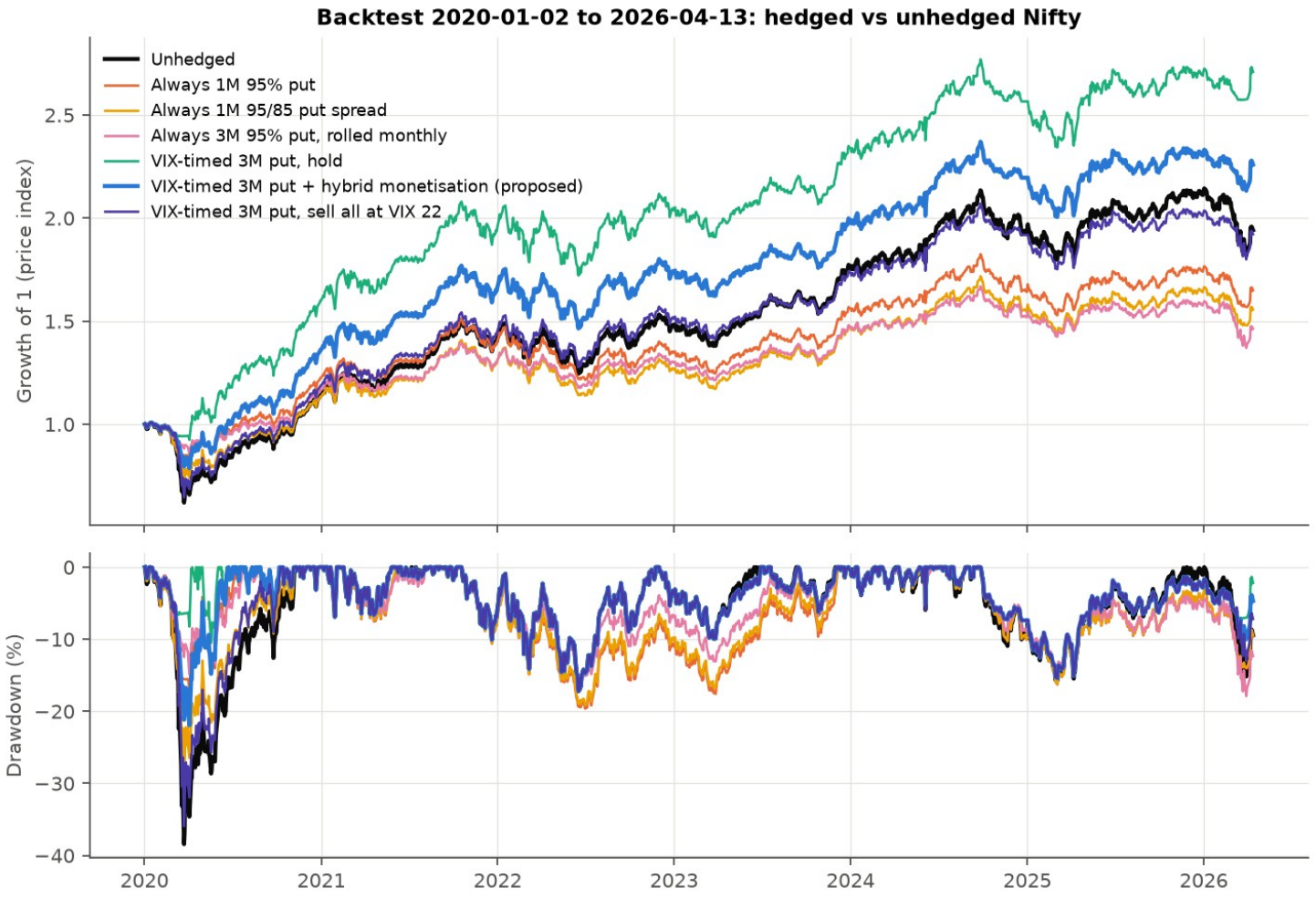
Fear is most overpriced after a spike

VRP of 4.0 pts in the high regime: the time to sell, not buy

Three months buys time

Slower decay and more vega than 1-month puts, and it survives a lag between signal and crash

Backtest: higher return, smaller falls



13.9% vs 11.1%

Annual return vs unhedged

−22% vs −38%

Max drawdown vs unhedged

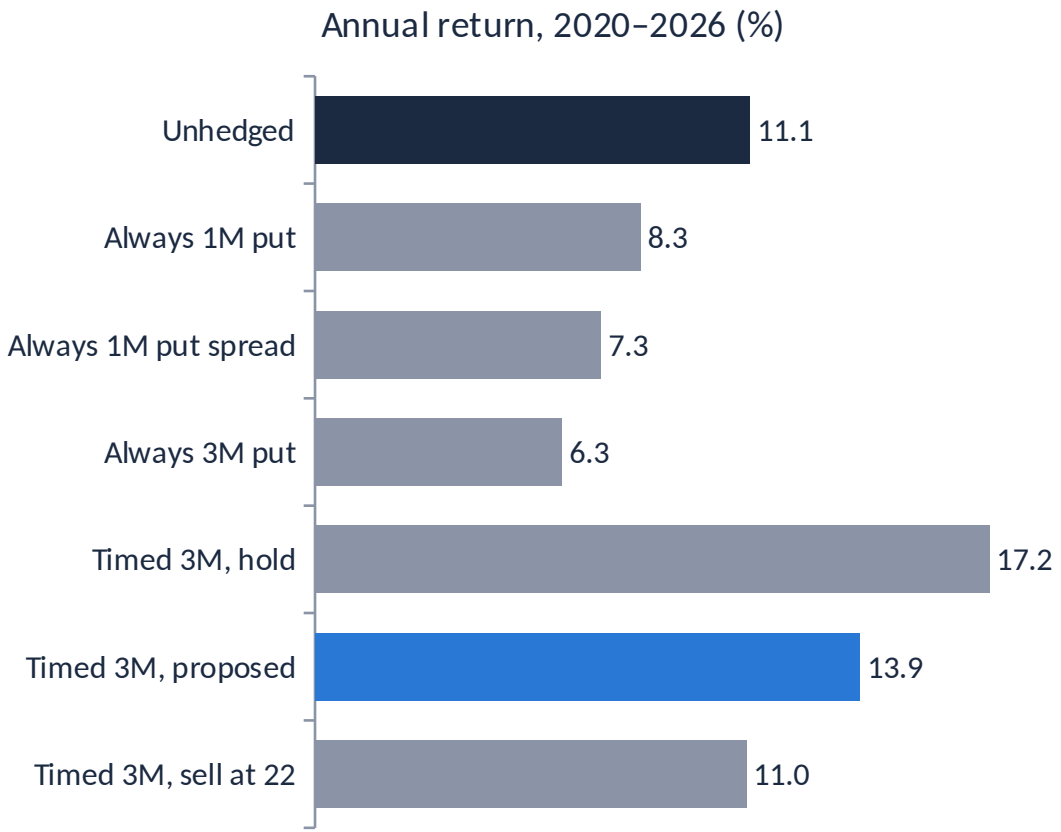
2.9% vs 7.5%

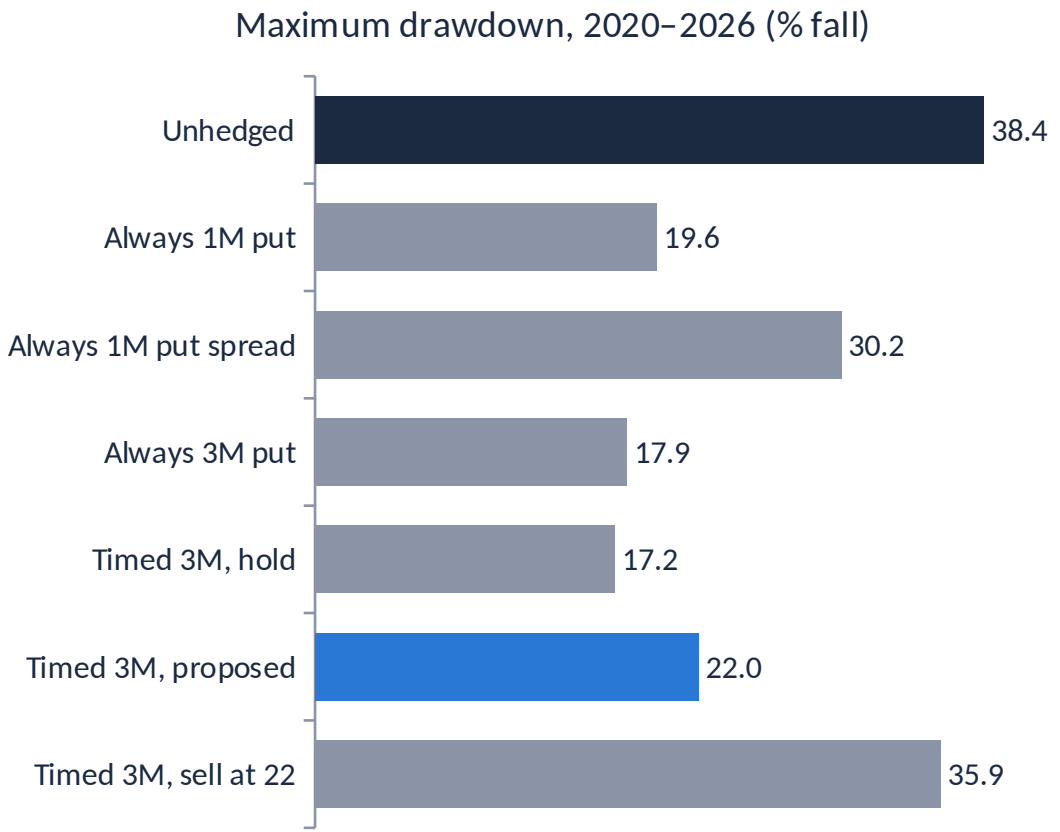
Premium a year vs always-on put

45%

Share of days holding puts

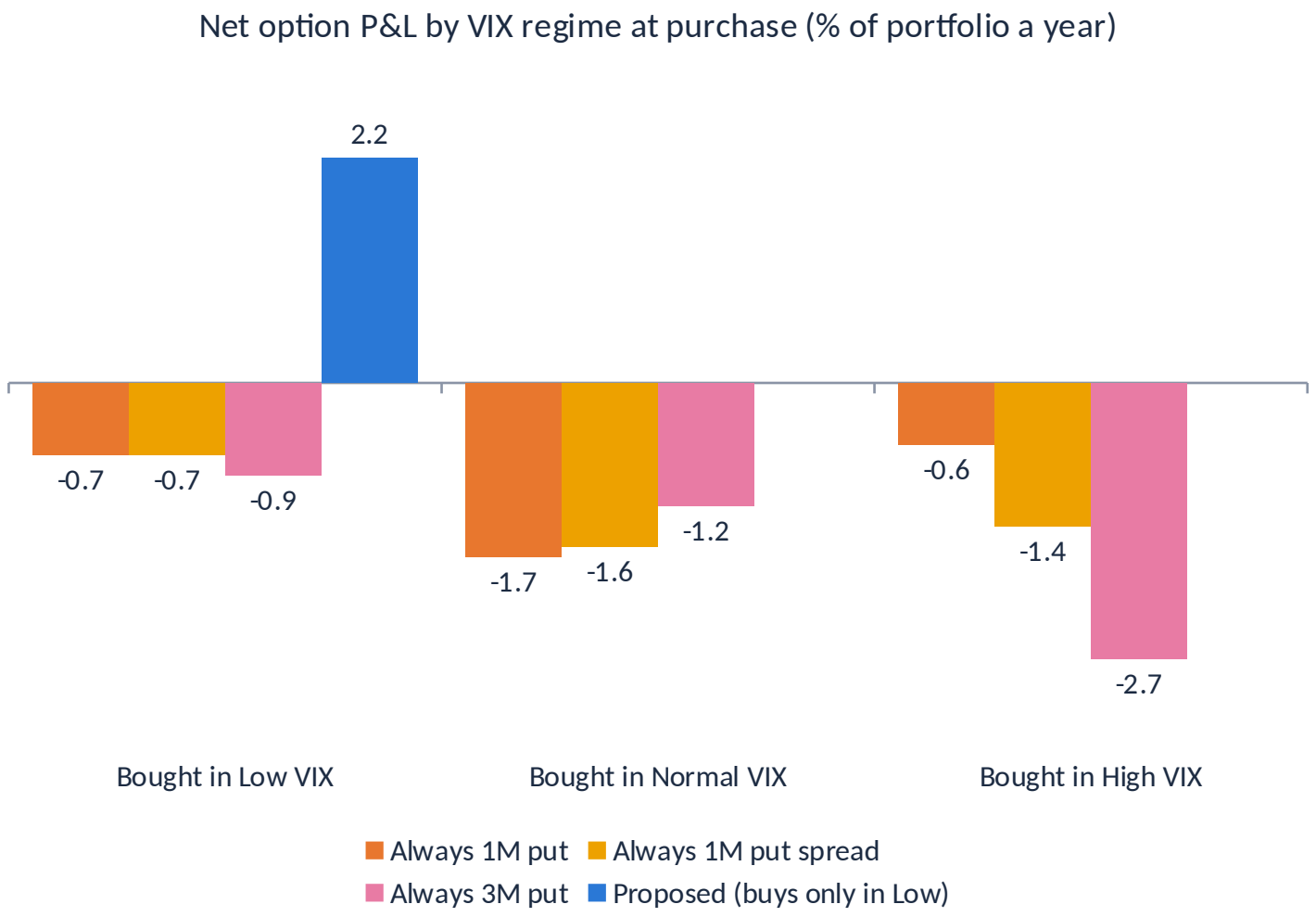
Always-on hedges cost more than they saved





Net option cost a year: 3.0% to 4.9% for always-on hedges, −4.4% to +0.2% for timed hedges (negative means the options returned more than they cost).

Hedges bought in calm markets paid



Proposed hedge vs unhedged, by the regime of each day

−2.1% a year
Low-VIX days: the insurance carry

+2.0% a year
Normal-VIX days

+6.3% a year
High-VIX days, when puts pay out

−9.2% a year
Always-on put on high-VIX days

Protection works from a calm start

Episode (portfolio return)	Unhedged	Always 1M put	Always 3M put	Proposed
COVID crash, 17 Jan to 23 Mar 2020	−38.4%	−15.5%	−12.4%	**−21.1%**
War and rate hikes, Jan to Jun 2022	−16.5%	−17.1%	−13.9%	**−16.5%**
Election result, 3 to 4 Jun 2024	−5.9%	−3.6%	−3.1%	**−5.2%**
Yen-carry unwind, 1 to 5 Aug 2024	−3.8%	−2.6%	−1.9%	**−1.9%**
Slow correction, Sep 2024 to Mar 2025	−15.8%	−16.3%	−14.7%	**−15.4%**
Tariff shock, 28 Mar to 7 Apr 2025	−5.8%	−4.3%	−3.4%	**−5.8%**

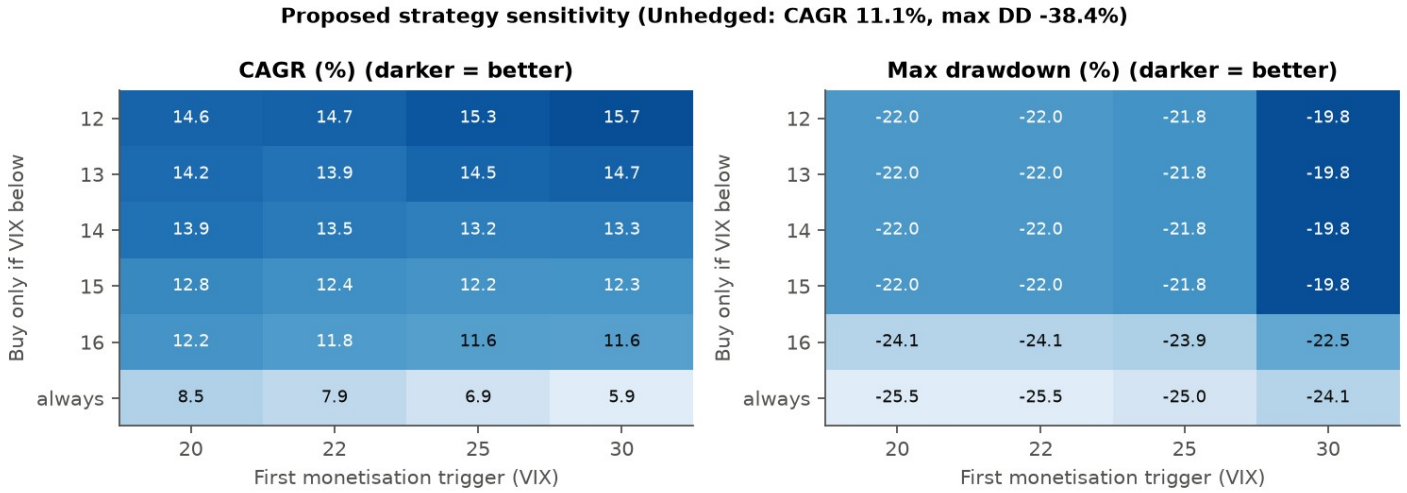
Works: shocks from a calm start

January 2020 and August 2024 began with VIX near or below 13, so a cheap put was already held

Fails: slow grinds, or VIX already rising

2022, late 2024 and April 2025 started with VIX in the normal band, so no hedge was bought

What the result depends on



Buying only below VIX 12 to 15 beat buying regardless of VIX in every cell: 12.2–15.7% a year against 5.9–8.5%, with drawdowns of about −20% to −22%.

Pricing check: with skew of 0 to 1 vol pt per 1% and half-spreads of 0 to 1 pt, the proposal returned 12.9–15.1% vs 6.0–11.1% for the always-on put.

Robust: buy only in low VIX

Holds across thresholds, skew and bid/ask assumptions

Fragile: tenor

1 to 2-month puts expired before the March 2020 crash: about 10.5% a year with −36% drawdown

Fragile: one crash

Jul 2020 to Apr 2026: 14.9% vs 15.4% unhedged, same −17.2% drawdown. Cost about 0.5% a year

Limitations

1

Modelled premiums

Priced from India VIX, not traded quotes. Real 3-month puts in calm markets probably cost more, because the term structure slopes up and skew is steeper.

2

Short history

India VIX data here starts in 2020: one true crash, and thresholds set on the same data they are tested on.

3

Synthetic expiries

Options expire on the first trading day of each month, not on NSE expiry days. The results show timing can matter a lot.

4

Costs left out

No dividends, taxes, STT, brokerage or margin. All trades at the daily close.

Buy protection cheap, sell it into fear

- Nifty option buyers pay about 2 vol points of risk premium, and about 4 points just after shocks
- India VIX is a good real-time gauge of fear and of option prices, not of market direction
- Event premiums build for weeks and collapse within a day, as on 4 June 2024
- A VIX-timed 3-month put costs about 0.5% a year in calm periods and cut the 2020–26 drawdown from 38% to 22%

Recommendation

Treat the VIX-timed put as cheap crash insurance that pays rarely, not as a source of return. Use 3-month tenors and sell half into the first spike.